COMP3018: Report (Literature Review & Programming Project)

Cognitive Robotics

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TODO

- [ ] verify word count is allowed.

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Vernon, Metta and Sandini (2007) – PDF NOT in papers/ folder; download from: https://www.robotcub.org/misc/papers/07_Vernon_Metta_Sandini_IEEE.pdf

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	298, 304	S1.1, S1.2	Vernon et al. (2007, p. TODO)	Try p. 155	Section "What is Cognition?"; cognition cycle diagram (anticipate, learn, adapt + perception/action)
- [ ]	306	S1.2	Vernon et al. (2007, p. TODO)	Try p. 163	Memory section; episodic vs semantic memory distinction (Tulving's taxonomy)

Sciutti et al. (2023) – `papers/Sciutti et al. (2023) - The Present and the Future of Cognitive Robotics.pdf`

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	298, 304, 314, 509	S1.1-S2.2	Sciutti et al. (2023, p. 160)	p. 160	"flexible, context-sensitive action, knowing what they are doing and why"; "reason about their actions and modify their behavior"
- [ ]	302	S1.2	Sciutti et al. (2023, p. 160) – "intersection of Robotics, AI, Cognitive Sciences"	p. 160	CHECK: this exact phrase may NOT be here; may come from Sandini, Sciutti & Vernon (2021) encyclopaedia entry instead
- [ ]	350	S1.5	Sciutti et al. (2023, pp. 162-163)	pp. 162-163	CHECK: "integrating machine learning techniques with model-based approaches" – may only be on p. 162, not spanning 163

Tapus, Matarić and Scassellati (2007) – `papers/Tapus, Matarić and Scassellati (2007) - Socially Assistive Robotics.pdf`

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	312	S1.3.1	Tapus et al. (2007, p. TODO)	Try p. 35	PARO listed: "robotic animal toys, such as a seal (i.e., PARO [2])"
- [ ]	509	S2.2	Tapus et al. (2007, p. 35)	p. 35	"helping human users through social rather than physical interaction"

Wada and Shibata (2007) – PDF NOT in papers/ folder

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	312, 342	S1.3.1, S1.4.2	Wada and Shibata (2007, p. 974)	p. 974	NEEDS MANUAL CHECK – download PDF; look for PARO reducing agitation / mood improvement in elderly

Fong, Nourbakhsh and Dautenhahn (2003) – **`papers/Fong, Nourbakhsh and Dautenhahn (2003) - A Survey of Socially Interactive Robots.pdf`**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	314	S1.3.1	Fong et al. (2003, p. 145)	p. 145	Section 1.2; Breazeal's four classes of social robots; "shallow models of social cognition" under Social Interface
- [ ]	326	S1.3.3	Fong et al. (2003, p. 149)	p. 149	Section 2.3 Embodiment; "mutual perturbation" / "perturbatory channels"
- [ ]	509, 538	S2.2, S2.3.3	Fong et al. (2003, p. 148)	p. 148	CHECK: p. 148 covers design issues, NOT emotion recognition. Try p. 155 (human-oriented perception listing) or p. 156 (speech/facial emotion analysis)

Lee and See (2004) – **`papers/Lee and See (2004) - Trust in Automation Designing for Appropriate Reliance.pdf`**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	318	S1.3.2	Lee and See (2004, p. 54)	p. 54	Definition in italics: "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability"

Hancock et al. (2011) – **`papers/Hancock et al. (2011) - A Meta-Analysis of Factors Affecting Trust in Human-Robot Interaction.pdf`**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	320, 334	S1.3.2, S1.4.1	Hancock et al. (2011, p. 522) – performance strongest predictor	p. 522	Table 1; "performance factors were more strongly associated (r = +0.34)"; Cohen's d = +0.71
- [ ]	318	S1.3.2	Hancock et al. (2011, p. 522) – "observation cannot reliably disambiguate"	p. 522	CHECK: this POMDP-style claim may NOT appear anywhere in Hancock et al.; possibly misattributed
- [ ]	334	S1.4.1	Hancock et al. (2011, p. 522) – "29 studies, modest variance"	pp. 520-522	29 studies stated on p. 520; overall r = +0.26 on p. 521; paper uses "moderate" not "modest"

Chen et al. (2020) – **`papers/Chen et al. (2020) - Trust-Aware Decision Making for Human-Robot Collaboration.pdf`**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	320	S1.3.2	Chen et al. (2020, p. 6)	p. 6 (article page :6)	Section 3.4 "Maximizing team performance"; Fig. 3 (Trust-POMDP model); "We maintain a belief b over the human's trust"

Garcez and Lamb (2023) – **papers/Garcez and Lamb (2023) - Neurosymbolic AI The 3rd Wave.pdf**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	320, 511, 663	S1.3.2, S2.2, S2.5	Garcez and Lamb (2023, p. 12389)	Local PDF is arXiv preprint (pp. 1-28); p. 12389 = journal page 3 of published version	CHECK against published *AI Review* version (journal pp. 12387-12406). On ~p. 3: third wave / neural-symbolic labour division

Nikolaidis, Hsu and Srinivasa (2017) – **papers/Nikolaidis, Hsu and Srinivasa (2017) - Human-Robot Mutual Adaptation in Collaborative Tasks.pdf**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	320, 624	S1.3.2, S2.3.3	Nikolaidis et al. (2017, p. 625)	p. 625	"69 samples"; "U = 180, p = 0.048"; MOMDP mutual-adaptation condition
- [ ]	334	S1.4.1	Nikolaidis et al. (2017, p. 627)	p. 627	MOMDP discussion; r = -0.066 (no correlation between trustworthiness and inferred adaptability)

Brooks (1991) – PDF NOT in papers/ folder; download from: https://people.csail.mit.edu/brooks/papers/rep

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	326	S1.3.3	Brooks (1991, p. TODO)	??	NEEDS MANUAL CHECK – find argument that intelligence emerges from physical interaction, not abstract representation

Matarić et al. (2007) – PDF NOT in papers/ folder; download from: https://pmc.ncbi.nlm.nih.gov/articles/P

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	326	S1.3.3	Matarić et al. (2007, p. TODO)	??	NEEDS MANUAL CHECK – find stroke survivors engaging more with embodied robot than screen-based alternative

Tapus, Ţăpuş and Matarić (2008) – PDF NOT in papers/ folder; download from: https://link.springer.com/article/10.1007/s11370-008-0017-4

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	326	S1.3.3	Tapus, Ţăpuş and Matarić (2008, p. TODO)	??	NEEDS MANUAL CHECK – find adaptive personality matching (interaction distance/speed) improving task performance

Papadimitriou and Tsitsiklis (1987) – **papers/Papadimitriou and Tsitsiklis (1987) - The Complexity of Markov Decision Processes.pdf**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	332	S1.4.1	Papadimitriou and Tsitsiklis (1987, p. 448)	p. 448	Theorem 6: "The partially observed problem is PSPACE-hard"

Pineau, Gordon and Thrun (2003) – **papers/Pineau, Gordon and Thrun (2003) - Point-Based Value Iteration An Anytime Algorithm for POMDPs.pdf**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	332	S1.4.1	Pineau et al. (2003, p. 1025)	p. 1025	First page / abstract: "Point-Based Value Iteration (PBVI) algorithm"

Kaelbling, Littman and Cassandra (1998) – **papers/Kaelbling, Littman and Cassandra (1998) - Planning and Acting in Partially Observable Stochastic Domains.pdf**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	332, 663	S1.4.1, S2.5	Kaelbling et al. (1998, p. 120)	p. 120	CHECK: p. 120 is the tiger problem (Section 5.1, toy example). For POMDP framework definition try p. 105; for belief-state planning try p. 108

Silver and Veness (2010) – **papers/Silver and Veness (2010) - Monte-Carlo Planning in Large POMDPs.pdf**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	332	S1.4.1	Silver and Veness (2010, p. 1)	p. 1	Abstract: "Monte-Carlo algorithm for online planning in large POMDPs"

Broadbent, Stafford and MacDonald (2009) – PDF NOT in papers/ folder; download from: https://link.springer.com/article/10.1007/s12369-009-0030-6

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	334	S1.4.1	Broadbent et al. (2009, p. TODO)	??	NEEDS MANUAL CHECK – find acceptance depending on matching robot behaviour to user expectations

Desai et al. (2013) – **papers/Desai et al. (2013) - Impact of Robot Failures and Feedback on Real-Time Trust.pdf**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	334, 538	S1.4.1, S2.3.3	Desai et al. (2013, p. 256)	p. 256	CHECK: local PDF is conference format with no printed page numbers. Verify against HRI 2013 proceedings. Look for: trust drops after reliability failures, slow recovery

Wachter, Mittelstadt and Floridi (2017) – **`papers/Wachter, Mittelstadt and Floridi (2017) - Why a Right to Explanation Does Not Exist in the GDPR.pdf`**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	340	S1.4.2	Wachter et al. (2017, p. TODO)	Try p. 76 (abstract) or p. 82 (Section 3 heading)	p. 76: "GDPR does not implement a right to explanation"; p. 82: Section heading "Why there is no 'right to explanation'"

Sharkey (2014) – **`papers/Sharkey (2014) - Robots and Human Dignity.pdf`**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	342	S1.4.2	Sharkey (2014, p. 6 (VERIFY))	Manuscript p. 6 = journal p. 68	"a robot that dealt impersonally with an older person, without knowing or using their name or their preferences…" in Nordenfelt Dignity of Identity context

Sharkey and Sharkey (2012) – **`papers/Sharkey and Sharkey (2012) - Granny and the Robots Ethical Issues in Robot Care for the Elderly.pdf`**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	350	S1.5	Sharkey and Sharkey (2012, p. 27)	p. 27	Abstract/introduction; concerns about reducing human contact. Detailed argument is on pp. 30-31

Ahn et al. (2022) – **`papers/Ahn et al. (2022) - Do As I Can Not As I Say Grounding Language in Robotic Affordances.pdf`**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	511	S2.2	Ahn et al. (2022, p. 1)	p. 1	Abstract: "constrain the model to propose natural language actions that are both feasible and contextually appropriate"

Smedegaard (2019) – **`papers/Smedegaard (2019) - Reframing the Role of Novelty within Social HRI from Noise to Information.pdf`**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	511, 663	S2.2, S2.5	Smedegaard (2019, p. 4)	p. 4 (= proceedings p. 414)	Novelty as "original feature of experience". CHECK: the novelty-fading engagement claim may be stronger on p. 2 (proceedings p. 412)

Ji et al. (2023) — **`papers/Ji et al. (2023) - Survey of Hallucination in Natural Language Generation.pdf`**

OK?	Line(s)	Section	Citation as written	Go to page…	You should see…
- [ ]	647	S2.3.4	Ji et al. (2023, p. 3)	p. 3	"deep learning based generation is prone to hallucinate unintended text"

- ☐ if enough time: ermengent cognitive robot architecture
- ☐ cite Iuliia Kotseruba1 · John K. Tsotsos1
- ☐ use 'inference' natrually, i.e., not so perfect that it is likely ai generated but instead slightly not 100% correct like a human would do, ygm?

Mentor

Scenario 1) patient takes meds = + 100 points Scenario 2) Patient is annoyed = -10 points When you implement tis you need to be careful to structure the rewards such that the system does not pester the patient as it seeks to maximise it's rewards You'll need to play around with the precise ratios between 1 (behaviour you want to encourage) and 2 (behaviour you want to discourage) to achieve the desired attitude from the robot

DR ALY QUESTIONS:

- ☐ so maths and code won't affect word count right? However I never explcitly knew if captions within a figure-diagram count towards word count?

First-to-do:

- ☐ **Most critical:** verify all page numbers and sentences manually
- ☐ **Most critical:** verify all links to papers manually
- ☐ do all page number TODOs
- ☐ implement loads of peer-reviewed papers everywhere again

General:

- ☐ ensure no overused complex words
- ☐ Download Broadbent via Plymouth library
- ☐ Run past Gemini
- ☐ decides the reward based on what it oberserves = inferring the reward
- ☐ discuss POMDP maths
- ☐ use wording from 3018 Task-4 Proposal Google Doc

- ☐ talk about the LfD (learning-from-demonstration)
- ☐ talk about IRL (inverse reinforcement learning)
- ☐ relatively talk about how it relates to others, motivation
- ☐ CRAMS figure verified – 5/6 actions appear (Back_Off absent because true state never sustained Low Trust long enough). Update report Task 4 discussion to: 1) explain why the action timeline shows context-sensitive selection (link each action cluster to the reward territory that produced it), 2) note Back_Off correctly absent given the Medium-trust initial state and stress profile, 3) highlight META-ADAPT triggers (red dotted lines) as evidence of metacognition detecting the stress event within 2 steps
- ☒ USE 'misclassified'
- ☐ consider a project wherein it is 'cogntive robotics' (lecture 9) ensure it involves what we have learnt in the labs
- ☐ write code like lecturer in: `3018-cw/learning/workshops/[X] emotional-speech-recognition/solution.py`
- ☐ 'persons'
- ☐ make the robot kinda like how I disucssed you should make it in the set exercises
- ☐ discuss mathematical notiation for the POMDP stuff??? (if not done in set exercises)
  - ☐ maths and diagrams affect wordcount?
  - ☐ Trust-POMPDP diagram
  - ☒ Cite: Chen, M., Nikolaidis, S., Soh, H., et al., "Trust-Aware Decision Making for Human-Robot Collaboration: Model Learning and Planning", ACM Trans. Hum.-Robot Interact., 9(2), 2020
  - ☐ model trust in lates diabram? or just latex the fundamental diagtam of POMDP similar to lecture slied in POMDP lectures (10,11)
  - ☐ latex diagram of continous state in POMDP similar to non-monotnoric graph (not about non-monitonic itsekf just a similar graph)
  - ☐ utilise lec-7 for POMDP insights; similar to machine learning set exercises where i give a quick breakdown if word count allowance
- ☐ args and kwargs (if i can do this)
- ☒ peer-reviewed or conference papers
- ☐ In this section, you should focus on providing enough description of the supervised learning, neural network, and naïve Bayes models.
- ☐ Do not assume the reader knows the basics. Dedicate specific paragraphs to explicitly defining the algorithms and the broader category (Supervised Learning) before diving into your implementation.
- ☐ Then, refer to some studies that have utilised neural networks and naïve Bayes models in your area using the selected database
- ☐ Ensure your literature review in the introduction explicitly cites papers that use your specific dataset (or very similar ones), establishing a clear baseline before you begin
- ☐ TODO: talk about the LfD (learn-from-demonstration)
  - ☐ TODO: talk about IRL (inverse reinforcement learning)
- ☐ TODO: talk about the LfD (learn-from-demonstration)
  - ☐ TODO: talk about IRL (inverse reinforcement learning)

LECTURE 9 – TODOs FOR ASSESSMENT 2 REPORT Task 3: Literature Review (Assistive Robotics Essay): - [X] Frame assistive robotics as requiring cognitive capabilities, not just social behaviour – use Aly's hierarchy: "intelligence deployed over the social layer, not vice versa" - [X] Reference Sciutti et al. (2023)

definition of cognitive robotics (replaced Cangelosi book – not peer-reviewed) - [X] Use Vernon, Metta and Sandini (2007) cognition cycle (replaced Vernon 2014 book – not peer-reviewed) - [X] Discuss theory of mind, prospection, and episodic memory as future directions/current gaps in assistive robotics - [X] Acknowledge the 42-definitions problem to argue the field still lacks consensus on what cognition actually is (shows critical thinking) - [X] Connect embodied cognition ("intelligence means body") to ethical implications of robots entering intimate care spaces Task 4: Programming Project - [ ] GET APPROVAL FROM ALY – confirm extending set exercises POMDP into cognitive architecture implementation is acceptable; confirm simulation-only is fine - [ ] Wait for Lectures 10/11 on cognitive architectures before finalising design – Aly said the building blocks reappear in those lectures - [ ] Explicitly map every system component to Aly's cognitive building blocks (perception, attention, action selection, memory, learning, reasoning, metacognition, prospection) - [ ] Add metacognition module (new from this lecture) – system monitors its own reasoning, flags when repeated actions produce negative outcomes - [ ] Implement explicit episodic vs semantic memory distinction – Aly specifically said "please distinguish or remember these two" - [ ] Label trust inference as theory of mind explicitly - [ ] Label POMDP planning over future states as prospection explicitly - [ ] Label observation filtering as attention (selective/suppressive) - [ ] Frame all actions as goal-directed (Aly's term for purposeful cognitive actions vs reactive behaviour) - [ ] Include cognitive architecture diagram in report showing how building blocks interconnect - [ ] Use the term "cognitive architecture" – Aly defined this as the system that "puts all these basic building blocks together and supports the communications between all" - [ ] Frame project as cognitive robotics (not just social robotics) – Aly said nobody has ever done this; "invitation for challenging minds" - [ ] Don't skip definitional rigour – define cognitive robotics, cognition, and key terms precisely in the Background section - [ ] 5-min video: walk through a scenario showing perceive -> attend -> reason -> act -> learn -> adapt cycle in action

Cross-check with Gemini.

1- Task (3) Literature Review

☒ [X] ## 1.1. Introduction

Ageing populations and a shrinking care workforce positioned **assistive robotics** *(human-supportive robots within physical, cognitive, and social realms of impairment affecting daily-living activities)*; a prominent technological response to a widening care gap. The field spans physical prosthetics; surgical assistance; neurodivergent support; exoskeletal rehabilitation, and social companionship; this essay however focuses on *socially* assistive robotics (*SAR*), an assistive-robot subfield wherein the robot "focuses on helping human users through social rather than physical interaction" (Tapus, Matarić and Scassellati, 2007, Abstract), as here most active research and ethical tension converge.

Robots now administer medication reminders, facilitate rehabilitation exercises, and therapeutically companionate in clinical and domestic settings: reduced caregiver burden, improved patient outcomes wherein residents became "more active and more communicative, both with each other and their caregivers" (Wada and Shibata, 2007, p. 973), and increased "social interaction" among elderly residents (Wada and Shibata, 2007, p. 972, Abstract).

However, most-current assistive robots operate at what Sciutti et al. (2023, p. 160) call the social layer: they react to immediate stimuli but lack the cognitive depth to anticipate user needs, remember past interactions, or reason about their own performance. Sciutti et al. argue that effective assistive robots should be *cognitive*: capable of "flexible, context-sensitive action, knowing what they are doing and why they are doing it." Vernon, Metta and Sandini (2007, p. 151) formalise this requirement via a "virtuous cycle that is embedded in an ongoing process of action and perception" (the agent anticipates → learns → adapts to achieve autonomy). This essay contends that assistive robotics should graduate from reactive social behaviour to cognitive capability (intelligence deployed *over* the social layer; not vice versa) if it is to deliver sustained, personalised support. The following sections survey the theoretical foundations thereof: evaluate prominent applications through this cognitive lens, discuss challenges and ethical implications, and identify future directions.

☐ ## 1.2. Literature Review

Cognitive robotics: defined by Sciutti et al. (2023, p. 160), lies at the intersection of Robotics, Artificial Intelligence, and Cognitive and Biological Sciences, combining "sensorimotor behaviour, higher-level functions, and social capabilities of an intelligent robot" (cf. Lecture 9, slides 4-5). This interdisciplinary grounding distinguishes it from conventional robotics *(treats the robot as purely engineered)* and from social robotics *(addresses interaction behaviour without necessarily modelling cognitive processes).* The distinction is consequential: a robot that smiles when a patient smiles is social; a robot that infers *why* the patient is smiling, and adjusts its future strategy accordingly, is *cognitive.

- ☐ verify following lecture slides

Vernon, Metta and Sandini (2007, p. TODO) synthesise the field's definitional plurality into a core cycle (cf. Lecture 9, slide 14). The European Network for Advancement of Artificial Cognitive Systems (euCognition) catalogued 42 definitions of cognition, yet the common thread therein is: anticipation, learning, and adaptation, intersected with perception and action to create autonomy (cf. Lecture 9, slide 13). This cycle provides an architectural checklist for assistive robots: a system that cannot direct its gaze toward relevant stimuli whilst suppressing irrelevant ones *(selective attention; Lecture 11, slide 19)*, anticipate the outcome of its actions *(prospection)*, learn from past interactions *(memory)*, or adapt its strategy when performance declines *(metacognition)* is, per this framework, not yet cognitive. Sciutti et al. (2023, TODO: verify p. 160) further specify that cognitive robots should "reason about their actions and modify their behavior to improve their effectiveness"; a capacity termed *theory of mind*, wherein the agent infers another's hidden mental state from observable behaviour.

Furthermore, memory is not monolithic. Vernon, Metta and Sandini (2007, p. TODO) distinguish *episodic memory (records of specific past experiences and their contextual outcomes)* from *semantic memory (general knowledge about the world, including spatial relationships and factual constraints).* For example, assistive-medication robots need episodic memory to recall that a user refused medication after a restless night, and semantic memory to know certain drugs cannot also be administered. Whilst the 42-definitions problem confirms the field lacks consensus on what cognition per se *is*, the common thread (anticipation, learning, adaptation) is precisely what assistive robotics demands.

- ☐ ## 1.3. Applications

1.3.1 Therapeutic and Emotional Support

The PARO therapeutic seal robot represents one of the most-widely deployed platforms within socially assistive robotics (Tapus, Matarić and Scassellati, 2007, p. TODO). Wada and Shibata (2007, p. 974) demonstrate that PARO reduces agitation and improves mood in patients with dementia, utilising tactile sensors and auditory processing to modulate its behaviour in response to touch and voice. Clinical trials report that urinary stress indicators "significantly improved" after PARO's introduction (Wada and Shibata, 2007, p. 978, Table II), thus the platform has been adopted in care homes across Japan, Europe, and the United States.

Notwithstanding these benefits, PARO operates at the reactive layer. It possesses no theory of mind (it cannot infer *why* a patient is agitated: loneliness, pain, confusion) nor episodic memory of what calmed this patient previously. A cognitively-equipped therapeutic robot, by contrast, would anticipate mood shifts via prospection, recall that music soothed this patient yesterday via episodic memory, and adapt its strategy via metacognition. Insofar as PARO's effectiveness plateaus because it cannot personalise its responses over time, the cognitive gap is not merely theoretical but potentially clinically consequential. Fong, Nourbakhsh and Dautenhahn (2003, p. 145) formalise this gap via Breazeal's taxonomy: PARO occupies the 'social interface' level (human-like cues but "shallow models of social cognition"), whereas Sciutti et al.'s (2023, p. 160) vision of robots "knowing what they are doing and why" demands the 'socially intelligent' level. The distance between these levels is the cognitive deficit assistive robotics should close.

1.3.2 Medication Adherence and Daily Living Support

Medication non-adherence imposes substantial costs on healthcare systems, and elderly patients with polypharmacy regimens are particularly vulnerable to missed or incorrect doses. Robots in this domain face a different challenge from therapeutic companionship: trust and cognitive load are latent variables that cannot be directly measured, only inferred from noisy behavioural proxies. Lee and See (2004, p. 54) define trust as "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability"; a definition foregrounding the unobservable nature that necessitates probabilistic modelling. A user may comply with a medication prompt despite low trust (e.g. time pressure), or indeed refuse despite high trust (e.g. task complexity), and thus the observation alone cannot reliably disambiguate the underlying state (Hancock et al., 2011, p. 522).

The Partially Observable Markov Decision Process (POMDP) provides formal machinery for this uncertainty. Chen et al. (2020, p. 6) demonstrate a Trust-POMDP wherein the robot maintains a belief distribution over trust and selects actions that maximise long-term collaboration, showing belief-space planning outperforms fixed strategies in the tested collaborative scenario. Garcez and Lamb (2023, p. 12389) identify the neuro-symbolic paradigm as the 'third wave' of AI, wherein neural subsystems (e.g. large language models) handle perception whilst symbolic subsystems (e.g. POMDPs) govern temporal reasoning, providing the temporal scaffold stateless systems lack. Nikolaidis, Hsu and Srinivasa (2017, p. 625) provide empirical corroboration: in a collaborative task ($n = 69$), robots utilising mutual adaptation via a Mixed Observability MDP (modelling human adaptability as a latent variable) were rated significantly more trustworthy than fixed-policy alternatives ($U = 180$, $p = 0.048$). This aligns with Hancock et al.'s (2011, p. 522) finding that robot performance attributes are the strongest trust predictors, whilst demonstrating that belief-space planning as advocated by Chen et al. (2020, p. 6) translates into measurable trust gains.

1.3.3 Physical Rehabilitation and Mobility

Robotic exoskeletons and assistive manipulators for stroke recovery and mobility support constitute a third application domain. These systems need to adapt in real time not only to the patient's physical state (joint angles and muscle activation patterns) but also to their psychological state: motivation, frustration, and fatigue are internal variables that determine whether a patient perseveres or disengages.

Embodied cognition becomes essential. Brooks (1991, p. TODO) argues that intelligence emerges from physical interaction with the environment rather than abstract representation, whereas Fong, Nourbakhsh and Dautenhahn (2003, p. 149) operationalise this as "perturbatory coupling": the more channels of mutual influence between robot and environment, the more embodied the system. A rehabilitation robot therefore occupies a uniquely cognitive niche, as it should sense the patient's body, reason about current capabilities, and adapt appropriately. A purely language-based or screen-based interface cannot achieve this; Matarić et al. (2007, p. TODO) confirm as much empirically, finding that stroke survivors engaged more enthusiastically with a physically embodied assistive robot than with screen-based alternatives. Tapus, Ţăpuş and Matarić (2008, p. TODO), in fact show that embodiment alone is insufficient: adaptive personality matching (adjusting interaction distance and speed to the user's traits) further improved task performance, suggesting rehabilitation robots require not just physical presence but cognitive adaptation to the individual. The cognitive building blocks required (haptic perception, prospective planning of difficulty, episodic memory of the patient's trajectory) thus suggest an embodied cognitive architecture is necessary rather than a disembodied controller.

☐ ## 1.4. Discussion

1.4.1 Challenges

Tapus, Matarić and Scassellati (2007, *.pdf*-p. 6) projected that by 2012 SAR systems would demonstrate "marked improvement in learning/training/recovery of the user"; yet PARO, the most-deployed platform nearly twenty years later, *still* cannot remember yesterday's session. Three challenges explain this stalled trajectory. Firstly, computational intractability: solving $POMDPs$ exactly is PSPACE-complete (Papadimitriou and Tsitsiklis, 1987, p. 448 {TODO VERIFY}), and the belief simplex grows exponentially with state-space dimensionality. Whilst approximate solvers such as point-based value iteration (Pineau, Gordon and Thrun, 2003, p. 1025; Kaelbling, Littman and Cassandra, 1998, p. 120) and online Monte-Carlo tree search (Silver and Veness, 2010, p. 1) mitigate this, real-time cognitive processing within embodied systems remains an open challenge, particularly when multiple unobserved variables (trust, load, emotion) require tracking simultaneously.

Secondly, the measurement problem: trust, cognitive load, and emotional state are not directly observable; observations thereof are noisy proxies at best. Hancock et al.'s (2011, p. 522) meta-analysis of 29 studies finds that even the strongest correlates of trust explain only modest variance, whilst Broadbent, Stafford and MacDonald (2009, p. TODO) note that acceptance itself depends on matching robot behaviour to user expectations rather than trust alone. Desai et al. (2013, p. 256) further demonstrate that trust dynamics are non-linear, building slowly through consistent performance but degrading rapidly after errors; and thus a single misclassified observation can cascade into inappropriate action selection. Nikolaidis, Hsu and Srinivasa (2017, p. 627), however, demonstrate that mutual adaptation partially mitigates this fragility: when the robot models human adaptability as a latent variable, trust persists even during strategy disagreements, suggesting the variance Hancock et al. report may stem from studies that treat the human as a static rather than co-adaptive partner.

Finally, adaptation without exploitation: a robot that runs inference on cognitive load could, in principle, time its medication requests to coincide with periods of high vulnerability, thereby maximising compliance at the expense of user autonomy. The reward function governing the POMDP's policy should therefore encode ethical constraints alongside clinical objectives, ensuring that the optimisation target is genuine adherence rather than coerced compliance.

1.4.2 Ethical Implications

Assistive robots operating in intimate care spaces (bedrooms, bathrooms, rehabilitation clinics) continuously collect sensitive behavioural data. Facial expressions, vocal patterns, and movement trajectories constitute biometric data, yet regulatory frameworks have not kept pace with deployment. Wachter, Mittelstadt and Floridi (2017, p. TODO) argue that even the General Data Protection Regulation provides no enforceable "right to explanation" of automated decisions; a gap particularly concerning in healthcare wherein recommendations directly affect patient wellbeing.

Moreover, over-reliance on assistive robots risks eroding functional independence. If a robot consistently anticipates and pre-empts needs via prospection, the user may disengage from self-directed activity, thereby creating a dependency that contradicts the assistive mandate. Sharkey (2014, p. 6 (VERIFY)) frames this via Nordenfelt's 'Dignity of Identity': "a robot that dealt impersonally with an older person, without knowing or using their name or their preferences would also be likely to negatively affect their feelings of dignity." This implies that only cognitively-equipped robots (those with episodic memory of individual users) can avoid dignity violations; reactive systems such as PARO, regardless of their therapeutic benefits (Wada and Shibata, 2007, p. 974), risk infantilisation precisely because they cannot personalise. The responsibility gap compounds this further: when a care robot administers incorrect medication, liability falls ambiguously between manufacturer, deployer, and clinician.

Furthermore, the deployment of assistive robots is not equitable: wealthy nations with sufficient infrastructure and research investment stand to benefit, whilst low-income populations face a widening digital divide in access to care technologies. The question of whether assistive robots displace human carers or augment them remains unresolved.

The ethical watchword is therefore proactive regulation: design-stage ethics that anticipate failure modes before deployment, rather than reactive patchwork after harm. Per the embodied cognition thesis, if intelligence indeed requires a body, and that body enters the most intimate spaces of vulnerable persons, then the ethical stakes of assistive cognitive robotics are uniquely high.

- ☐ ## 1.5. Conclusion

Assistive robotics stands at an inflection point. Current systems (PARO, medication prompt robots, rehabilitation aids) deliver measurable benefits within narrow operational envelopes, yet their reactive architectures limit sustained, personalised effectiveness. The Vernon, Metta and Sandini (2007) cognition cycle provides the architectural blueprint for graduating beyond this plateau: assistive robots that anticipate (prospection), remember (episodic and semantic memory), reason about others' mental states (theory of mind), and monitor their own performance (metacognition) would constitute a qualitative advance over the most-capable systems deployed.

The neuro-symbolic paradigm offers a viable path toward this vision, as the Trust-POMDP framework attests (Chen et al., 2020). Sciutti et al. (2023, pp. 162-163) independently identify the integration of learning with model-based approaches as cognitive robotics' most-prominent trajectory; that this converges with Garcez and Lamb's (2023, p. 12389) 'third wave' thesis from AI theory suggests the direction is robust rather than parochial. Future applications will likely extend beyond single-task assistance toward cognitively autonomous home-dwelling companions: robots that proactively monitor health indicators, anticipate daily needs via episodic memory, and adapt their interaction style to the user's evolving cognitive and emotional state. Sharkey and Sharkey (2012, p. 27) identify this trajectory whilst cautioning that such systems risk replacing rather than supplementing human-care, and therefore the field should pursue cognitive capability and ethical governance in concert. Figure~1 visualises this trajectory. The ultimate test, per the embodied cognition thesis, is a robot that can sense, remember, anticipate, and adapt within the physical world, whilst respecting the autonomy and dignity of the persons it serves.

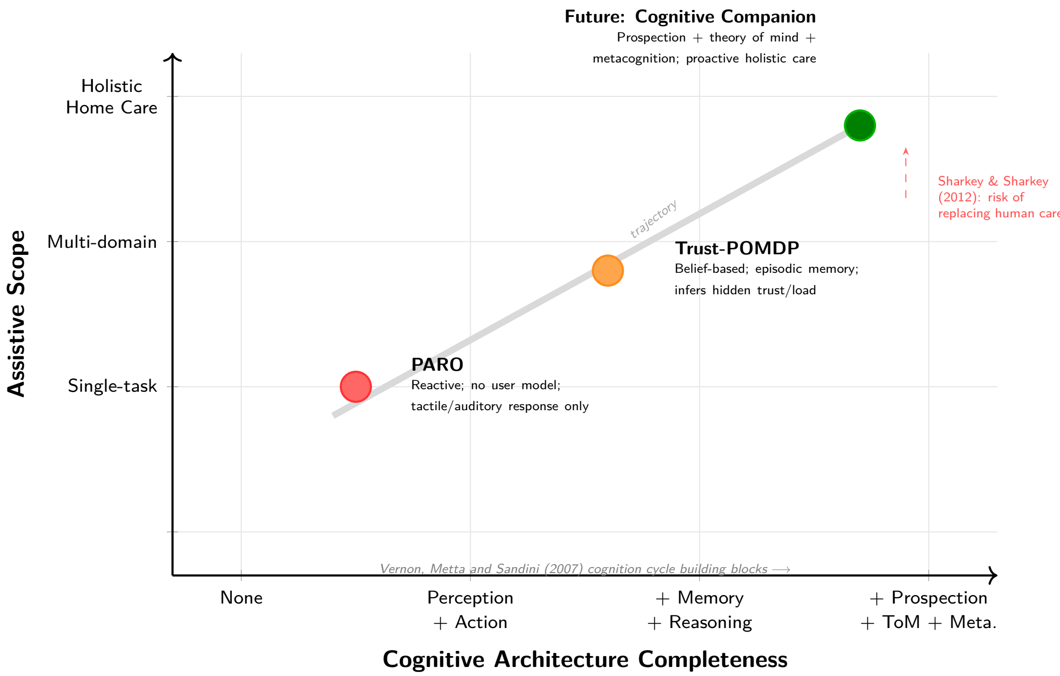

Figure 1: Trajectory of assistive robotics from reactive single-task systems (PARO) through adaptive belief-based architectures (Trust-POMDP) toward cognitively autonomous home-dwelling companions. Expanding assistive scope without expanding cognitive capability is insufficient; the diagonal trajectory requires both.

1.6. Task-3 References

1.6.1 Papers and Journal Articles

- ☐ verify all links work and extract each papers insights

REMOVE: one checkmark for working link second for insight gathered

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2- Task (4) Noval Programming Project

2.1. Introduction (10%)

2.2. Background (10%; Alfie)

GAZE sits within socially assistive robotics *(the deployment of robots to support users through social interaction rather than physical contact)*; Tapus, Matarić and Scassellati (2007, p. 35) define this sub-field as systems that "assist users through social interaction," which distinguishes it from physically assistive platforms such as exoskeletons. Most-current platforms in this domain react to a single input signal (Fong, Nourbakhsh and Dautenhahn, 2003, p. 148), and thus suffer from the single-signal problem: a facial-expression classifier alone misreads a resting face as displeasure, whilst a response-time metric alone mistakes thoughtful deliberation for disengagement. Calvo and D'Mello (2010, p. 28) identify "the inherent challenges with unisensory affect detection," observing that the field remains dominated by "mostly unimodal approaches" despite the recognised advantage of multi-channel fusion; Poria et al. (2017, p. 99) further report that multimodal systems were "consistently (85% of systems) more accurate than their best unimodal counterparts, with an average improvement of 9.83%." This demands multi-signal fusion (the system weighs complementary channels together instead of trusting any one alone).

GAZE's core contribution is multi-signal emotional inference: the system simultaneously captures facial expression via a pre-trained CNN (Workshop 10), vocal emotion via an MLP speech-emotion classifier (Workshop 8), response time, and answer correctness via speech-to-text combined with LLM-based answer verification. The engine fuses these four raw inputs alongside three derived temporal signals (rolling correctness, consecutive silences, consecutive-wrong streaks) into a single inferred user-state governing all downstream adaptation. This architecture relies on a hybrid approach: the neural subsystem (OpenAI GPT-4.1) generates games and dialogue, whilst a symbolic rule-based engine (the AdaptiveEngine) governs state inference and difficulty adjustment via interpretable rules. Grounding LLM output in Pepper's physical affordances *(what the robot can actually do: speak, gesture, illuminate LEDs)* follows the affordance-aware principle advocated by Ahn et al. (2022, p. 1), wherein language models are constrained to actions the robot can perform. Smedegaard (2019, p. 4) further warns that initial engagement with social robots often reflects novelty rather than sustained interest; GAZE's adaptive game-switching mechanism directly targets this novelty-decay problem by responding to inferred disengagement with variety rather than repetition.

2.3. Methods & Setup (35%; Alfie)

2.3.1 System Architecture

GAZE operates across four sequential layers: 1) INPUT captures four simultaneous signals from the user (facial expression, vocal emotion, response time, answer correctness); 2) PROCESS fuses these via the AdaptiveEngine to infer the user's true emotional-cognitive state; 3) GENERATE constructs a dynamic prompt and dispatches it to OpenAI GPT-4.1 for game and dialogue generation; 4) OUTPUT delivers speech, gestures, and LED feedback on Pepper concurrently via threading. Computationally intensive processing (CNN inference, OpenAI API calls, adaptive-engine logic) runs on the laptop, whilst Pepper handles only physical I/O.

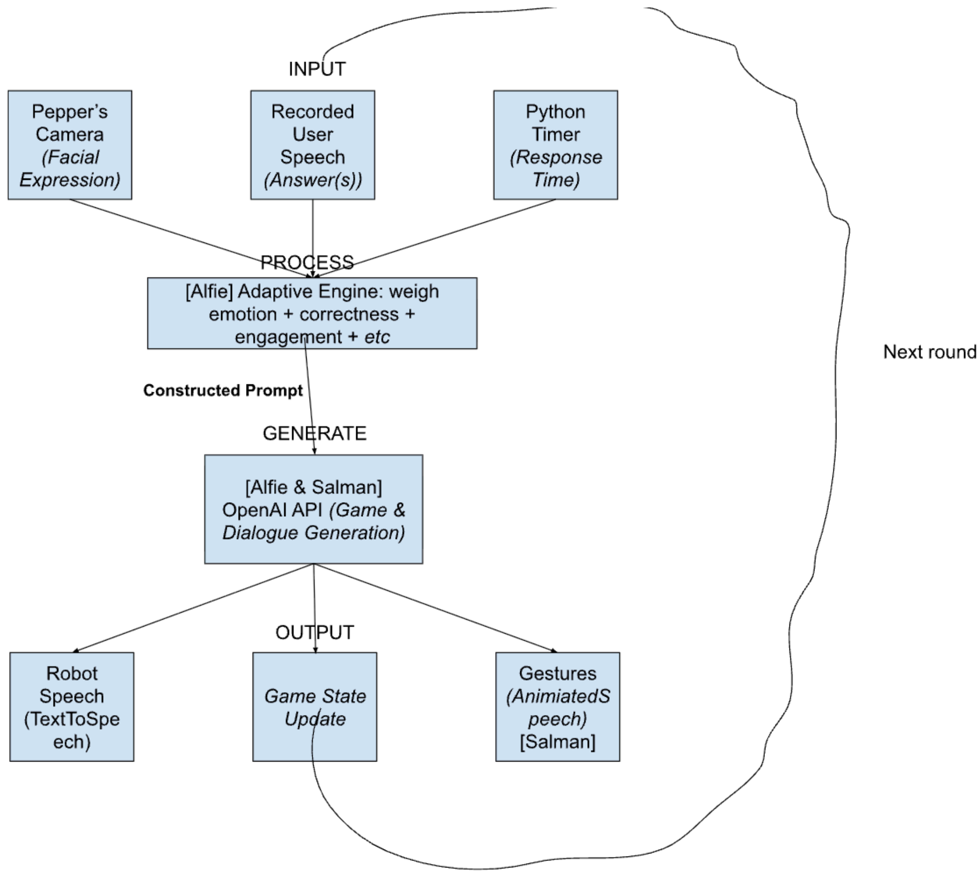

Figure 2: GAZE system architecture (adapted from the project proposal). Four input channels (facial expression, vocal emotion, response time, answer correctness) feed the AdaptiveEngine (PROCESS), which infers user-state and constructs a dynamic prompt for OpenAI GPT-4.1 (GENERATE); the resultant dialogue, gesture tag, and game-state update are delivered via Pepper's speech, motor, and LED subsystems (OUTPUT) concurrently. The loop circulates to the next round, now adapted.

2.3.2 Input Layer: Four Simultaneous Signals

1- Facial Expression (vision-based). The previously trained CNN from Workshop 10 classifies the user's expression into one of seven categories (Angry, Disgust, Fear, Happy, Neutral, Sad, Surprise) from a 48×48 greyscale face region. These categories build upon Ekman and Friesen (1971, pp. 127-128), whose cross-cultural results (Tables 1-2) show that "particular facial behaviors are universally associated with particular emotions," finding that even members of a preliterate (without written language) culture with "minimal opportunity to have learned to recognize uniquely Western facial expressions" identified the same six emotions; with the addition of a neutral baseline by modern datasets, this taxonomy remains "still the most popular perspective for FER" (Li and Deng, 2020, p. 1), albeit the known limitations regarding cultural bias and resting-face misclassification, which GAZE mitigates via cross-modal override rather than trusting any single frame. Pepper's camera captures the image; the largest detected face is selected and fed through the workshop preprocessing pipeline.

2- Verbal Answer (speech-based). Pepper records audio via `ALAudioRecorder` with dynamic silence detection calibrated to the room's ambient noise level at startup. Recording terminates when 1.5 seconds of silence follows detected speech, or a 12-second hard ceiling is reached. The recorded WAV is transcribed via OpenAI Whisper (`whisper-1`).

3- Vocal Emotion (audio-based). The same recorded WAV is passed through a pre-trained MLP speech-emotion classifier (Workshop 8) *before* transcription, classifying the speaker's vocal state into one of four emotions (calm, happy, fearful, disgust). The classifier extracts MFCC, chroma, and mel-spectrogram features; El Ayadi, Kamel and Karray (2011, p. 577) identify MFCCs as "the most promising features" for speech-emotion recognition, encoding the spectral envelope that shifts measurably between emotional states (e.g. raised pitch and energy in anger versus compressed range in sadness). This provides a second, independent modality for emotional inference; the camera captures a static frame whilst the voice captures the full spoken response, and thus the two modalities may legitimately disagree, wherein the decision logic arbitrates via performance data.

4- Response Time (engagement-based). A Python timer measures elapsed time from question delivery to recording completion, isolating user deliberation time from downstream API latency; the Whisper call occurs *after* the timer halts. A correct answer delivered after 25 seconds of deliberation indicates a fundamentally different user-state from one delivered in 3, and thus warrants different adaptive behaviour.

2.3.3 Process Layer: Multi-Signal State Inference

The AdaptiveEngine's `infer_state()` method weighs seven signals simultaneously to classify the user into one of five states: *Thriving*, *Comfortable*, *Struggling*, *Frustrated*, or *Disengaged*. The four raw inputs (facial expression, vocal emotion, response time, current-round correctness) are supplemented by three derived temporal signals: rolling correctness over the last five rounds, consecutive silence count, and consecutive wrong-answer streak length. The classification rules combine these signals in cross-modal ways: if the camera reads *Angry* but the user answers quickly and correctly, the engine resolves to *Comfortable (merely a resting face)*; if voice reads *happy* and rolling correctness exceeds 80%, the engine deduces *Thriving* regardless of facial expression; if both modalities agree on a negative state and correctness falls below the 40% floor, the engine infers *Frustrated*; and if voice reads *calm* whilst the camera reads a frown but correctness is adequate, the cross-modal override resolves to *Comfortable*, preventing a single misleading frame from triggering unnecessary intervention. Fong, Nourbakhsh and Dautenhahn (2003, p. 148) identify "emotion recognition" as a key capability for socially interactive robots, yet most implementations rely on a single modality; GAZE's multi-signal approach, wherein temporal streaks override or corroborate instantaneous readings, addresses the brittleness Desai et al. (2013, p. 256) observe when systems depend on singular, noisy observations. The thresholds (correctness floor of 0.4, ceiling of 0.8, response-time baseline of 30 seconds, consecutive-wrong trigger of 3) were derived from pilot testing: a correctness floor below 0.4 produced false positives during normal difficulty fluctuations, whilst a consecutive-wrong threshold below 3 triggered frustration interventions prematurely; conversely, thresholds above these values delayed intervention past the point of sustained negative affect.

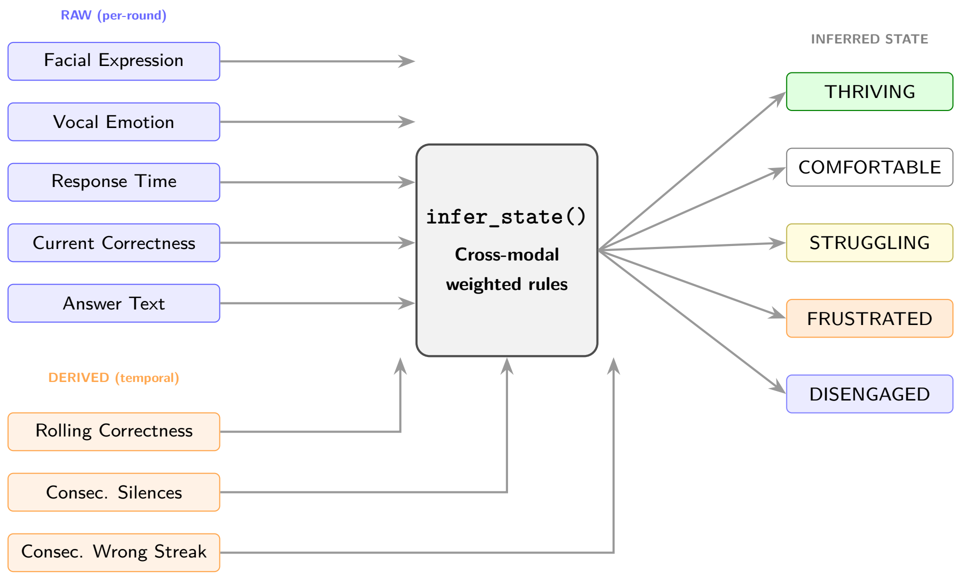

Figure 3: Multi-signal inference pipeline. Four raw inputs captured each round and three derived temporal signals computed from session history feed into `infer_state()`, which applies cross-modal rules (e.g. camera reads *Angry* but user answers fast and correctly → *Comfortable*; voice reads *calm* whilst camera frowns → cross-modal override to *Comfortable*) to classify the user into one of five states. Neither visual nor vocal modality is trusted in isolation.

The cross-modal inference rules are shown in Listing~1 (extracted from `gaze.py`, *lines 343–438*):

Listing 1: Multi-signal inference rules (excerpt from `gaze.py`).

```

# thriving: performing well regardless of resting face

if (correctness >= CORRECTNESS_CEILING
        and response_time < RESPONSE_TIME_BASELINE * 0.5):
    return InferredState.THRIVING

# camera says Angry but fast + correct → they're fine

if expression == "Angry" and correct
    and response_time < RESPONSE_TIME_BASELINE * 0.6:
    return InferredState.COMFORTABLE

# disengaged: multiple signals pointing to checked-out

if self.consecutive_silences >= SILENCE_THRESHOLD:
    return InferredState.DISENGAGED
if (expression == "Neutral"
        and response_time > RESPONSE_TIME_BASELINE
        and correctness < 0.5):
    return InferredState.DISENGAGED

# frustrated: struggling + negative expression

if expression in ("Angry", "Disgust")
    and correctness < CORRECTNESS_FLOOR:
    return InferredState.FRUSTRATED
if self.consecutive_wrong >= 3
```

```
    and expression in ("Angry", "Sad", "Fear"):
    return InferredState.FRUSTRATED
```

The `decide()` function then maps the inferred state to concrete adaptive actions: difficulty adjustment (easy, medium, hard), game switching (numbers ↔ letters when the user is frustrated or disengaged), hint provision, encouragement, and tone selection. This maps onto the mutual-adaptation paradigm identified by Nikolaidis, Hsu and Srinivasa (2017, p. 625), wherein the robot adjusts its strategy based on an evolving model of the user rather than following a fixed policy. Table~1 summarises the full state-to-action mapping.

Inferred State	Difficulty	Game Switch	Hint	Tone	LED Colour
Thriving	↑ ramp up	No	No	Energetic	Green
Comfortable	↑ if >70%	No	No	Neutral	White
Struggling	↓ ease off	No	Yes	Encouraging	Yellow
Frustrated	↓↓ Easy	After 4 wrong	Yes	Calm	Orange
Disengaged	—	After 2 silent	No	Energetic	Blue

Table 1: State-to-action mapping. The adaptive engine translates each inferred state into a specific combination of difficulty adjustment, game-type switch, hint provision, tone selection, and LED colour. Arrows indicate direction of difficulty change relative to the current level.

2.3.4 Generate Layer: Dynamic Prompt Construction

The prompt sent to GPT-4.1 is never static. `build_game_prompt()` assembles it fresh every round from the current tone instruction, live metrics (rolling correctness, response time, recent expressions and vocal emotions, inferred state), and adaptive instructions (hints, encouragement, game-switch directives). Separate OpenAI calls handle game generation and answer verification at different temperatures; the deterministic verification call mitigates the hallucination risk Ji et al. (2023, p. 3) identify in LLM-generated output by treating correctness as a classification task. A strict 10-second timeout wraps every API call; if the university network drops or OpenAI stalls, the robot falls back to a graceful dialogue acknowledging the failure and proceeds to the next round, thereby ensuring Pepper remains responsive rather than freezing mid-interaction.

2.3.5 Output Layer: Aligned Multimodal Response

Speech, gestures, and LED state fire concurrently via threading. Context-aligned gestures (e.g. animated speech for celebratory moments) execute alongside dialogue, whilst LED colours reflect the inferred state, providing a secondary non-verbal feedback channel.

2.3.6 Adaptation Self-Evaluation and Session Persistence

After each round, `evaluate_adaptation()` assesses whether the previous adaptation worked by comparing concrete outcome pairs (e.g. did a difficulty decrease produce a correct answer? did a game switch transition the user from a negative to a positive state?). The evaluation is injected into the next prompt as an `ADAPTATION FEEDBACK` block, so the LLM can adjust its dialogue accordingly; the result is a system that learns whether its adaptations are effective and communicates that to the generation layer. Session progress is saved to `gaze_save.json` after every round via progressive save, protecting data against unexpected crashes and supporting session resumption.

2.4. Outcome & System Analysis (30%)

2.5. Conclusion (10%; Alfie)

GAZE implements multi-signal emotional inference across two independent modalities *(facial expression via WS-10 CNN and vocal emotion via WS-08 MLP)* alongside response time, answer correctness, and derived temporal signals, which yields a more robust user-state estimate than any single channel in isolation. The adaptive engine translates the inferred state into concrete actions and evaluates whether those actions worked; this self-evaluation loop feeds back into the LLM prompt, creating a system that genuinely adapts rather than executing a fixed interaction script. The hybrid architecture (neural generation paired with symbolic rule-based inference) and the adaptive game-switching mechanism directly targets the novelty-decay problem Smedegaard (2019, p. 4) identifies. The multi-signal approach could transfer beyond game-based engagement to stroke rehabilitation re-engagement (Matarić et al., 2007), educational tutoring, or therapeutic support for neurodivergent users. Future work could replace the hand-coded classification thresholds with learned parameters derived from longitudinal user data, and fine-tune both models on in-session Pepper captures to improve classification accuracy within the specific deployment environment.

2.6 Task-4 References (5%)

☐ verify all references

Alfie's

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2.7. Appendix

2.7.1. Code

2.7.2. Video Demo

- YouTube link: test

Appendices

Appendix A: AI Declaration

Solo Work	**S1 - Generative AI tools have not been used for this assessment.**
Assisted Work	**A1 – Idea Generation and Problem Exploration** Used to generate project ideas, explore different approaches to solving a problem, or suggest features for software or systems. Students must critically assess AI-generated suggestions and ensure their own intellectual contributions are central.
	A2 - Planning & Structuring Projects AI may help outline the structure of reports, documentation and projects. The final structure and implementation must be the student's own work.
	A3 – Code Architecture AI tools maybe used to help outline code architecture (e.g. suggesting class hierarchies or module breakdowns). The final code structure must be the student's own work.
	A4 – Research Assistance Used to locate and summarise relevant articles, academic papers, technical documentation, or online resources (e.g. Stack Overflow, GitHub discussions. The interpretation and integration of research into the assignment remain the student's responsibility.
	A5 - Language Refinement Used to check grammar, refine language, improve sentence structure in documentation not code. AI should be used only to provide suggestions for improvement. Students must ensure that the documentation accurately reflects the code and is technically correct.
	A6 – Code Review AI tools can be used to check comments within the code and to suggest improvements to code readability, structure or syntax. AI should be used only to provide suggestions for improvement. Students must ensure that the code accurately reflects their knowledge and is technically correct.
	A7 - Code Generation for Learning Purposes Used to generate example code snippets to understand syntax, explore alternative implementations, or learn new programming paradigms. Students must not submit AI-generated code as their own and must be able to explain how it works.
	A8 - Technical Guidance & Debugging Support AI tools can be used to explain algorithms, programming concepts, or debugging strategies. Students may also help interpret error messages or suggest possible fixes. However, students must write, test, and debug their own code independently and understand all solutions submitted.
	A9 - Testing and Validation Support AI may assist in generating test cases, validating outputs, or suggesting edge cases for software testing. Students are responsible for designing comprehensive test plans and interpreting test results.
	A10 - Data Analysis and Visualization Guidance AI tools can help suggest ways to analyse datasets or visualize results (e.g. recommending chart types or statistical methods). Students must perform the analysis themselves and understand the implications of the results.
	A11 - Other uses not listed above Please specify:
Partnered Work	**P1 - Generative AI tool usage has been used integrally for this assessment** Students can adopt approaches that are compliant with instructions in the assessment brief. Please Specify:

Figure 4: Student Declaration of AI Tool use in this Assessment Table

I declare that I've used the AI tools listed below whilst preparing this assessment. I've read and understood the University of Plymouth's policy on the use of AI tools in assessment and confirm that my use falls within the coursework's allowed categories, i.e. **A2 (Planning and Structuring Projects)** and **A4 (Research Assistance)**.

AI Tool Used	**Purpose of Use**	**Extent of Use**
ChatGPT	Brainstorming project ideas and structuring the report **(A2)**	Initial brainstorming and final outline stages
ChatGPT	Reviewing structural alignment against grading criteria and mapping word-count budgets **(A2)**	After and midway through section-drafting
ChatGPT	Finding relevant pages to read in the paper **(A4)**	Few times if the paper is too long
ChatGPT	General conversations via web-search AI about prevalent papers to read about how the topics relates to others' studies **(A4)**	Few times at the end

- ☒ I understand that the ownership and responsibility for the academic integrity of this submitted assessment falls with me, the student.
- ☒ I confirm that all details provided above are an accurate description of how AI was used for this assessment.